Foreword

This book is about a revolution in our view of space and time, and its remarkable consequences, some of which are still being unraveled. It is also a fascinating account, written by someone closely involved, of the struggles and eventual success in a search for an understanding of what are possibly the most mysterious objects in the Universe-black holes.

It used to be thought obvious that the surface of the Earth was flat: It either went on forever or it had some rim that you might fall over if you were foolish enough to travel too far. The safe return of Magellan and other round-the-world travelers finally convinced people that the Earth's surface was curved back on itself into a sphere, but it was still thought self-evident this sphere existed in a space that was flat in the sense that the rules of Euclid's geometry were obeyed: Parallel lines never meet. However, in 1915 Einstein put forward a theory that combined space and time into something called spacetime. This was not flat but curved or warped by the matter and energy in it. Because spacetime is very nearly flat in our neighborhood, this curvature makes very little difference in normal situations. But the implications for the further reaches of the Universe were more surprising than even Einstein ever realized. One of these was the possibility that stars could collapse under their own gravity until the space around them became so curved that they cut themselves off from the rest of the Universe. Einstein himself didn't believe that such a collapse could ever occur, but a number of other people showed it was an inevitable consequence of his theory.

The story of how they did so, and how they found the peculiar properties of the black holes in space that were left behind, is the subject of this book. It is a history of scientific discovery in the making, written by one of the participants, rather like *The Double Helix* by James Watson about